Automated PPE Detection Using CNN-Based Object Detection

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1. INTRODUCTION

1.1 Problem Statement

The rapid urbanization and infrastructural development in Angeles City, Pampanga, have led to a surge in commercial construction and healthcare facility expansions. Within these high-risk environments, the Department of Labor and Employment (DOLE) strictly mandates the use of Personal Protective Equipment (PPE) to protect the labor force. However, enforcing consistent PPE compliance across dynamic worksites remains a persistent challenge.

Historically, automated safety compliance relied on sensor-based technologies such as Radio Frequency Identification (RFID) and Global Positioning Systems (GPS). While accurate, these methods require substantial initial investments, high maintenance, and force workers to wear invasive, uncomfortable tracking devices [2]. Conversely, traditional manual monitoring by safety officers is labor-intensive, reactive, and susceptible to human error. Local environmental factors, such as extreme tropical heat and high humidity, frequently induce fatigue in both workers and inspectors, increasing the likelihood of compliance lapses.

This project addresses the operational gap in safety enforcement by developing a vision-based automated PPE detection system utilizing the YOLOv8 object detection architecture. By continuously analyzing visual data from worksite cameras, the system can simultaneously detect multiple classes of safety gear without requiring physical sensors on the workers. This approach transforms standard video feeds into active monitoring tools, allowing safety personnel to efficiently enforce DOLE regulations and mitigate occupational hazards without increasing administrative overhead.

1.2 Scope

The scope of this research is strictly confined to worksites within Angeles City, Pampanga, specifically targeting active commercial building construction sites and medical facilities.

The proposed convolutional neural network (CNN) will be trained exclusively to detect three critical PPE classes:

1. Hard Hats (Safety Helmets): The standard requirement for overhead hazard protection.
2. High-Visibility Safety Vests: Essential for worker visibility near heavy machinery.
3. Face Masks: A strict regulatory requirement in medical buildings and particulate-heavy zones.

The system will utilize the YOLOv8 architecture to output bounding box coordinates and classification confidence scores. The project is restricted to 2D image analysis and excludes the detection of smaller items (e.g., safety goggles, gloves) due to standard camera resolution constraints. The final deliverable is intended as an assistive monitoring tool for safety officers, rather than a fully autonomous enforcement system.

1.3 Success Metrics

The performance of the proposed PPE detection system will be evaluated using standard object detection metrics. The following metrics will be used to assess the accuracy and efficiency of the system:

- **Mean Average Precision (mAP@0.5):** The primary metric for evaluating object detection accuracy, representing the average of the Average Precision (AP) across all classes at an Intersection over Union (IoU) threshold of 0.5. It is calculated as:

$$mAP = \frac{1}{N}\sum_{i=1}^{N} AP_i$$

- **Precision and Recall:** Precision quantifies the proportion of positive identifications that are correct (accuracy of the predictions), while Recall measures the proportion of actual positive instances correctly identified by the model (ability to find all targets).
- **Precision–Recall (PR) Curve:** A graphical representation that plots Precision against Recall at various threshold settings, illustrating the trade-off between the two metrics and demonstrating the model's reliability across different confidence levels.
- **Latency / Inference Time:** Measures how quickly the model processes an image. This metric helps determine whether the system can support real-time detection.
- **F_1Score:** The harmonic mean of precision and recall, providing a balanced evaluation of model robustness:

$$F_1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

1.4 Constraints

The development and evaluation of this PPE detection system face several practical and technical constraints:

- **Dataset Bias and Generalization**: Public PPE datasets often suffer from severe class imbalance. For example, these contain thousands of images of hard hats but very few of safety vests. They also tend to lack environmental diversity. Because of this, the model may struggle to adapt to the specific lighting, weather conditions, and demographic realities of worksites in Angeles City.
- **Hardware Limitations**: Training deep learning object detectors requires significant GPU power. Because we have restricted access to high-performance computing, we cannot train an overly complex model. Instead, we must rely on lightweight architectures like YOLOv8 and carefully optimize our batch sizes to avoid out-of-memory errors.
- **Time Constraints**: Model training, tuning, and testing must all be completed within a strict academic schedule. This tight timeline means we cannot use exceptionally large datasets or run exhaustive, multi-week training sessions.
- **Environmental Noise**: Real-world construction footage is messy. Lighting changes constantly, motion blur is common, and dirty clothing drastically reduces feature visibility [8]. Furthermore, workers are frequently hidden behind scaffolding, heavy machinery, or each other [5], [6]. This visual noise degrades image quality and makes it much harder for the model to draw accurate bounding boxes around the equipment.
- **Ethical and Privacy Considerations:**: Analyzing images of workers naturally raises privacy and surveillance concerns. To maintain strict ethical standards, this project is designed solely to detect safety gear. The system will not use facial recognition algorithms or store any personally identifiable information (PII).

2. RELATED WORKS

The transition from sensor-based safety monitoring (e.g., RFID tags) to vision-based computer vision systems has significantly reduced deployment costs and physical invasiveness on worksites [2]. Central to this transition is the advancement of deep learning, specifically the YOLO (You Only Look Once) architecture family, which formulates object detection as a single regression problem optimized for real-time inference [4], [7].

Recent empirical studies heavily favor the YOLOv8 architecture for construction safety due to its anchor-free detection head and superior capability in identifying small, distant objects. Al-Khiami and ElHadad [1] demonstrated the efficacy of YOLOv8 in detecting helmets, vests, and safety shoes, achieving a 95% mAP and emphasizing AI's potential in occupational health. Similarly, Biswas and Hoque [3] utilized YOLOv8 to concurrently detect PPE and heavy machinery, noting its robust performance even in limited lighting conditions.

However, raw detection architectures struggle with the complex realities of active worksites, particularly class imbalance and occlusion. Utomo et al. [12] addressed the underrepresentation of "no-helmet" and "no-vest" categories by applying Mixup and Mosaic data augmentation techniques, optimizing YOLOv8 hyperparameters to improve mAP@0.50 from 0.867 to 0.921. To further prevent model overfitting and handle environmental noise (such as dirty clothing and motion blur), Seth and Sivagami [9] proposed the integration of Test Time Augmentation (TTA) alongside image transformations like Histogram Equalization, Gamma Correction, and Contrast Stretching.

The practical deployment of these architectures necessitates operational efficiency on constrained edge hardware. A comparative study by Shubqizam et al. [10] evaluated YOLOv7 and YOLOv8 for real-time PPE monitoring on edge devices. While both models performed exceptionally well, the study highlighted the critical trade-off between detection accuracy and inference speed (FPS) required for live video monitoring. Furthermore, advanced methodologies are being developed to counter severe occlusion, such as utilizing context-guided data augmentation and lightweight CNN backbones to maintain high accuracy without exorbitant computational costs [5], [11].

3. DATASET PLAN

3.1 Source & Preprocessing

This study will utilize publicly available Personal Protective Equipment (PPE) detection datasets which will be sourced from open repositories such as Kaggle and Roboflow. These sources comprise annotated images of workers in construction and industrial environments. The datasets contain bounding box annotations necessary for supervised object detection training, specifically targeting hard hats, high-visibility vests, and face masks. Prior to model training, the raw imagery will undergo standard preprocessing, which will include resolution normalization (e.g., resizing to 640x640 pixels) and the conversion of bounding box coordinates into the required YOLOv8 format.

3.2 License

The selected datasets are distributed under open-source licenses designated strictly for academic and research purposes. To ensure strict compliance with the providers' licensing terms, the raw data

will not be hosted directly within the project repository. Instead, an automated data extraction script will be provided to allow for reproducible access directly from the original sources.

3.3 Consent

The imagery within these public datasets was crowdsourced or captured explicitly for research applications and does not contain clear Personally Identifiable Information (PII), such as readable name tags or high-resolution facial biometrics. Consequently, the use of this data for academic experimentation will align perfectly with standard ethical guidelines regarding worker privacy and informed consent.

3.4 Representativeness & Augmentation

While the baseline dataset captures various industrial environments, it inherently lacks representation of the specific geographic and environmental nuances of worksites in Angeles City, Pampanga. To mitigate this dataset bias and improve model generalization, robust data augmentation techniques will be applied during the training phase. Advanced transformations—including Mosaic and Mixup augmentations, alongside gamma correction, random scaling, and brightness adjustments—will be utilized to artificially simulate severe occlusion, dynamic worker positions, and the harsh tropical lighting conditions typical of the target deployment areas.

4. CNN COMPONENTS AND MVP

4.1 CNN Component

The core of the proposed system will be a one-stage Convolutional Neural Network (CNN) object detector, specifically the YOLOv8 architecture, optimized for recognizing personal protective equipment. Unlike traditional two-stage models, YOLOv8 performs simultaneous object localization and classification within a single forward pass, making it highly efficient for real-time inference. The network will be fine-tuned on the preprocessed worksite dataset, leveraging bounding box annotations to learn the distinct spatial features of hard hats, face masks, and high-visibility vests under varying environmental conditions.

4.2 Minimum Viable Product (MVP)

The Minimum Viable Product (MVP) will be an end-to-end, reproducible machine learning pipeline capable of automated PPE compliance monitoring. The deliverable will comprise three primary components:

- **Trained Inference Model:** A fine-tuned YOLOv8 weights file (e.g., best.pt) explicitly optimized for the three target PPE classes.
- **Evaluation Module:** An automated script that will compute and visualize overall performance metrics, including the precision-recall curve, mAP@0.5, and inference latency.
- **Inference Visualizer:** A processing script that will accept raw worksite imagery as input and output the predicted image, complete with overlaid bounding boxes and classification confidence scores.

As an advanced stretch objective, the system will demonstrate real-time inference capabilities by processing a live video or webcam feed.

5. ETHICS RISK REGISTER

1. Privacy and Data Handling. The continuous visual monitoring of workers raises valid privacy concerns regarding data collection and storage. To mitigate this risk, the training phase will rely exclusively on anonymized, publicly available datasets. Furthermore, the operational prototype will process video feeds inferentially which means, it will analyze frames for PPE compliance in real time without capturing, storing, or transmitting personally identifiable information (PII).

2. Surveillance Misuse. There is an ethical risk that safety-oriented computer vision systems could be repurposed for excessive workplace surveillance, such as micromanaging worker productivity or tracking break times. To prevent this misuse, the model will be narrowly engineered to recognize only specific safety gear. Clear operational guidelines will dictate that the system functions strictly as a safety auditing tool, not as a mechanism for behavioral tracking.

3. Algorithmic Inaccuracy and False Negatives. The object detection model is susceptible to incorrect predictions, particularly in poorly lit or highly occluded worksite environments. A false negative—failing to detect a worker missing a helmet—could leave that individual exposed to fatal hazards. To mitigate this, rigorous confidence thresholds will be applied to the model's outputs. The system will be explicitly designed as an assistive alert mechanism, requiring verification by a human safety officer before any operational interventions occur.

6. TEAM ROLES AND MILESTONES

6.1 Team Roles

The project team will be composed of up to four members, with each member assigned a specific role to ensure efficient development and coordination of the system.

Project Lead / Integration. The Project Lead will oversee the overall project scope, timeline, and coordination among team members. This role is responsible for integrating different system components, managing the GitHub repository, handling release tagging, and organizing the final project defense.

Data & Ethics Lead. The Data and Ethics Lead will handle dataset sourcing, licensing verification, and preprocessing of the PPE dataset. This role will also ensure that ethical considerations such as privacy, consent, and responsible data usage are addressed in the project documentation.

Modeling Lead. The Modeling Lead will focus on the design and implementation of the object detection model. This includes selecting the CNN architecture (e.g., YOLOv5, YOLOv8, or SSD), training the model, performing experiments, and conducting error analysis to improve detection performance.

Evaluation & MLOps Lead. The Evaluation and MLOps Lead will handle model evaluation and reproducibility. Responsibilities include calculating evaluation metrics such as mAP, precision, and recall, managing experiment logs, maintaining the training environment, and ensuring that the project can be reproduced using scripts and documentation.

6.2 Milestones

The development of the project will follow a structured timeline to ensure progress throughout the semester.

Week 1 – Proposal & Setup.

During this stage, the project topic will be finalized and the proposal will be completed. The GitHub repository will be initialized with the required project structure, including the README file, environment configuration, dataset download script, and project board.

Week 2 – Data Preparation & Initial Model Training.

The PPE dataset will be collected and preprocessed. Initial experiments will be conducted using the selected object detection model. Baseline results will be generated to evaluate the initial performance of the system.

Week 3 – Model Improvement & Final Integration.

The object detection model will be refined and evaluated using the selected metrics. Results, visualizations, and error analysis will be prepared. The final report, slides, and demonstration of the PPE detection system will also be completed.

7. REFERENCES

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